

Image Processing (303105381)

Mr. Anurag Kewat

Lecturer
Computer science and Engineering



UNIT-7

Image Segmentation

Outline

- **Image Segmentation**
- Detection of Discontinuities
- Edge Linking and Boundary Detection
- Thresholding
- Region-Based segmentation

Outcome of this unit:

After completing the students will be able to:

- To understand the fundamental concepts of Image Segmentation.
- Apply the different image segmentation techniques.
- Learn recognition techniques.

Image Segmentation

- **Segmentation divides an image into its constituent regions or objects.**
- **Segmentation of images is a difficult task in image processing. Still under research.**
- **Segmentation allows to extract objects in images.**
- **Segmentation is unsupervised learning.**
- **Model based object extraction, e.g., template matching, is supervised learning.**

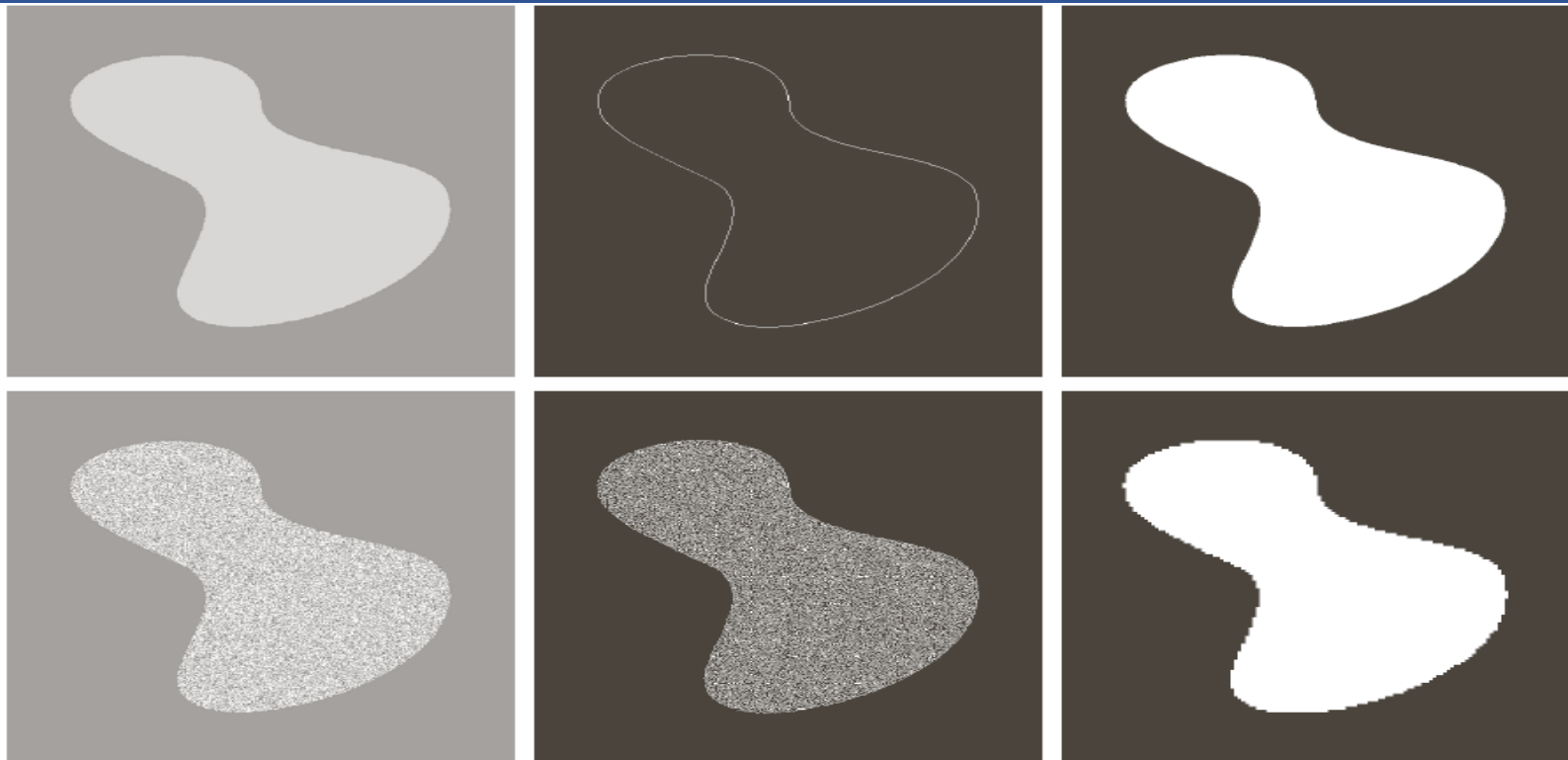
Segmentation Algorithms

- Segmentation algorithms are based on one of two basic properties of color, gray values, or texture: discontinuity and similarity.
- First category is to partition an image based on abrupt changes in intensity, such as edges in an image.
- Second category are based on partitioning an image into regions that are similar according to a predefined criteria. Histogram thresholding approach falls under this category.

fundamentals

- Let R represent the entire spatial region occupied by an image. Image segmentation is a process that partitions R into n sub-regions, $R_1, R_2, \dots, R_n$, such that





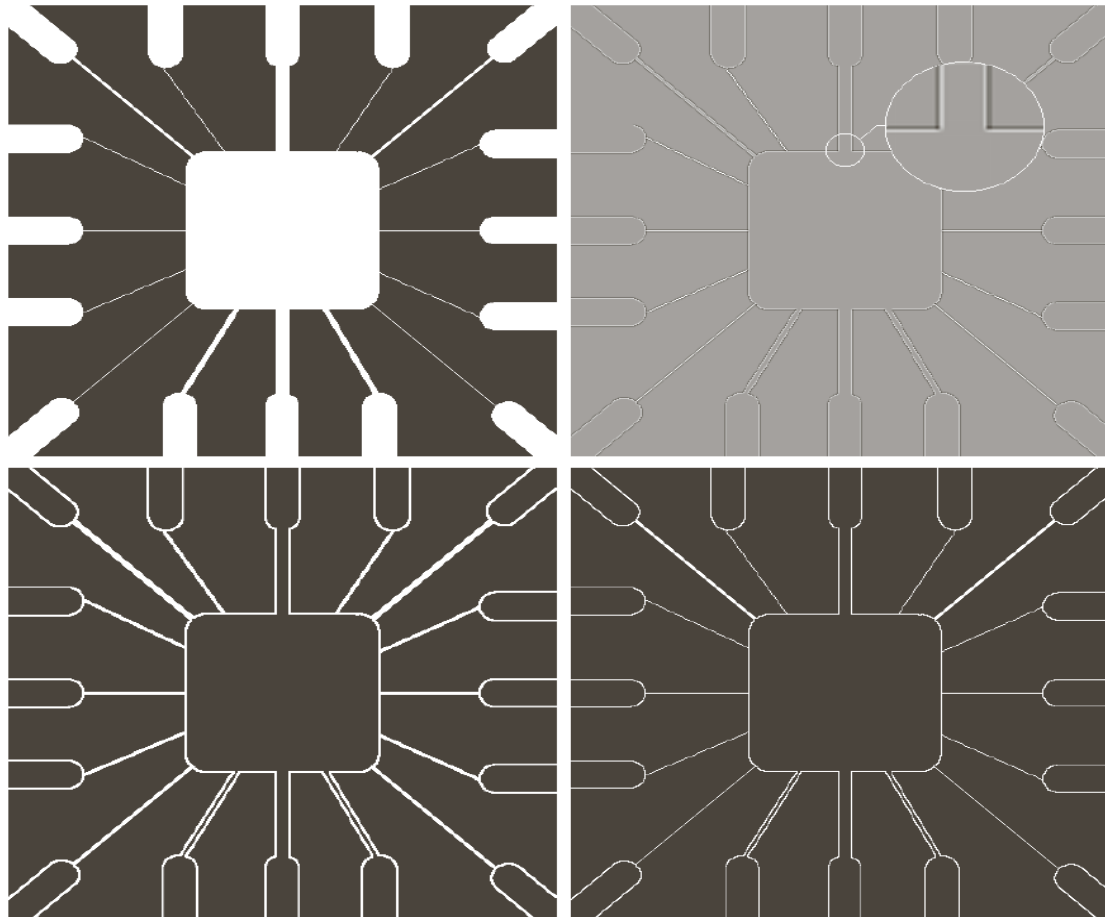
a	b	c
d	e	f

FIGURE 10.1 (a) Image containing a region of constant intensity. (b) Image showing the boundary of the inner region, obtained from intensity discontinuities. (c) Result of segmenting the image into two regions. (d) Image containing a textured region. (e) Result of edge computations. Note the large number of small edges that are connected to the original boundary, making it difficult to find a unique boundary using only edge information. (f) Result of segmentation based on region properties.

Line detection

- **Second derivatives to result in a stronger response and to produce thinner lines than first derivatives.**
- **Double-line effect of the second derivative must be handled properly**

continue



a	b
c	d

FIGURE 10.5

(a) Original image.
(b) Laplacian image; the magnified section shows the positive/negative double-line effect characteristic of the Laplacian.
(c) Absolute value of the Laplacian.
(d) Positive values of the Laplacian.

Detecting Line in Specified Directions

-1	-1	-1	2	-1	-1	-1	2	-1	-1	-1	2
2	2	2	-1	2	-1	-1	2	-1	-1	2	-1
-1	-1	-1	-1	-1	2	-1	2	-1	2	-1	-1
Horizontal			+45°			Vertical			-45°		

FIGURE 10.6 Line detection masks. Angles are with respect to the axis system in Fig. 2.18(b).

- Let R_1 , R_2 , R_3 , and R_4 denote the responses of the masks in Fig. 10.6. If, at a given point in the image, $|R_k| > |R_j|$, for all $j \neq k$, that point is said to be more likely associated with a line in the direction of mask k .

Edge Detection

- Edges are pixels where the brightness function changes abruptly

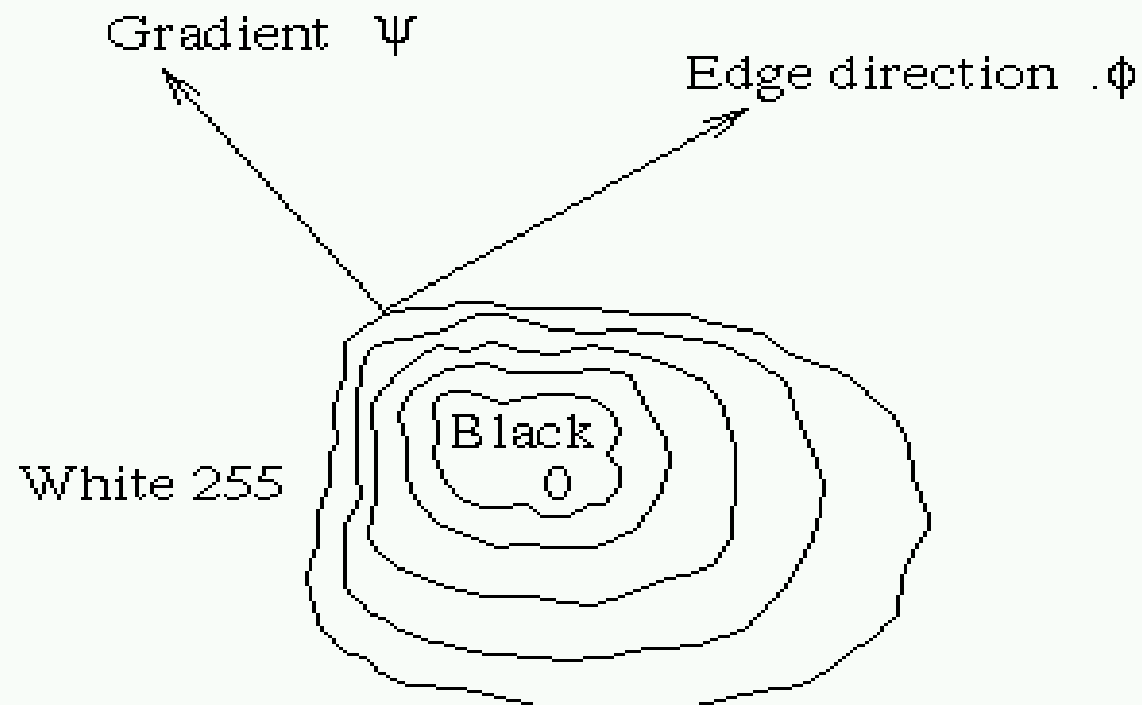


FIGURE 10.8
From left to right,
models (ideal
representations) of
a step, a ramp, and
a roof edge, and
their corresponding
intensity profiles.

Basic Edge Detection by Using First-Order Derivative

PU





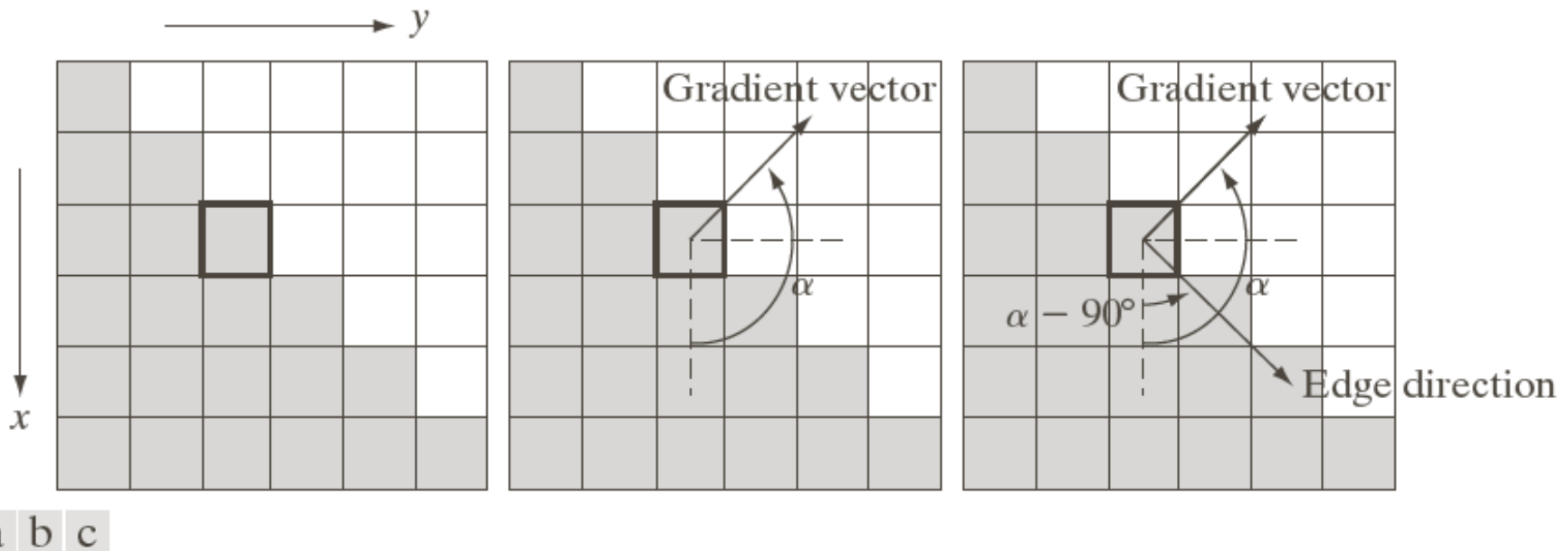


FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square in the figure represents one pixel.



-1
1

-1	1
------	-----

a b

FIGURE 10.13
One-dimensional
masks used to
implement Eqs.
(10.2-12) and
(10.2-13).



z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

	a	
b		c
d		e
f		g

FIGURE 10.14
A 3×3 region of an image (the z 's are intensity values) and various masks used to compute the gradient at the point labeled z_5 .

-1	0	0	-1
0	1	1	0

Roberts

-1	-1	-1	-1	0
0	0	0	-1	0
1	1	1	-1	0

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

0	1	1	-1	-1	a	b
-1	0	1	-1	0	c	d
-1	-1	0	0	1		

FIGURE 10.15
Prewitt and Sobel
masks for
detecting diagonal
edges.

Prewitt

0	1	2	-2	-1	0
-1	0	1	-1	0	1
-2	-1	0	0	1	2

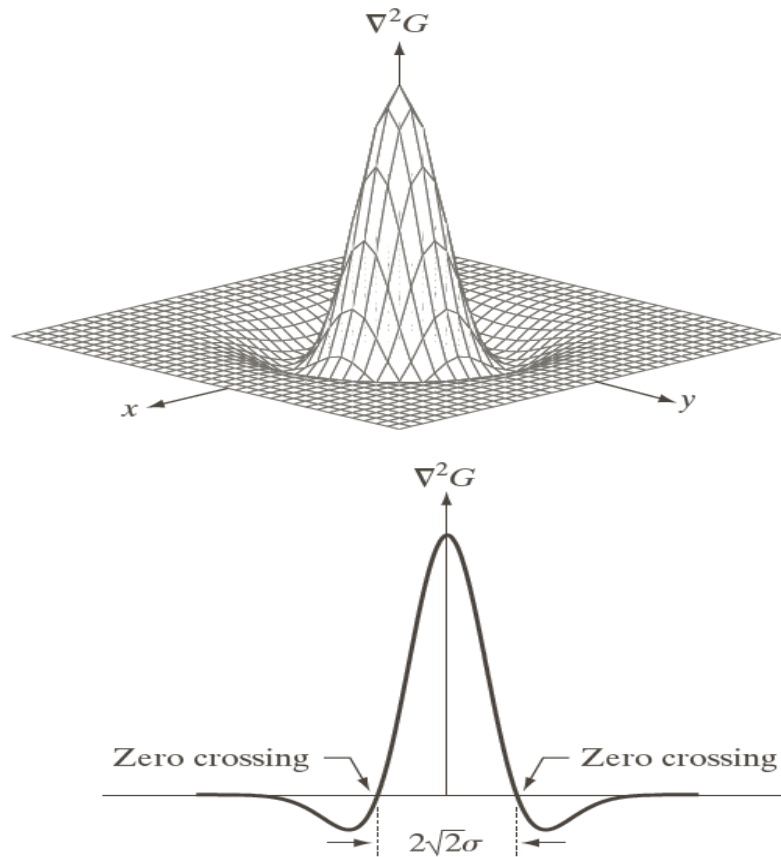
Sobel

Advanced Techniques for Edge Detection

PU



PU

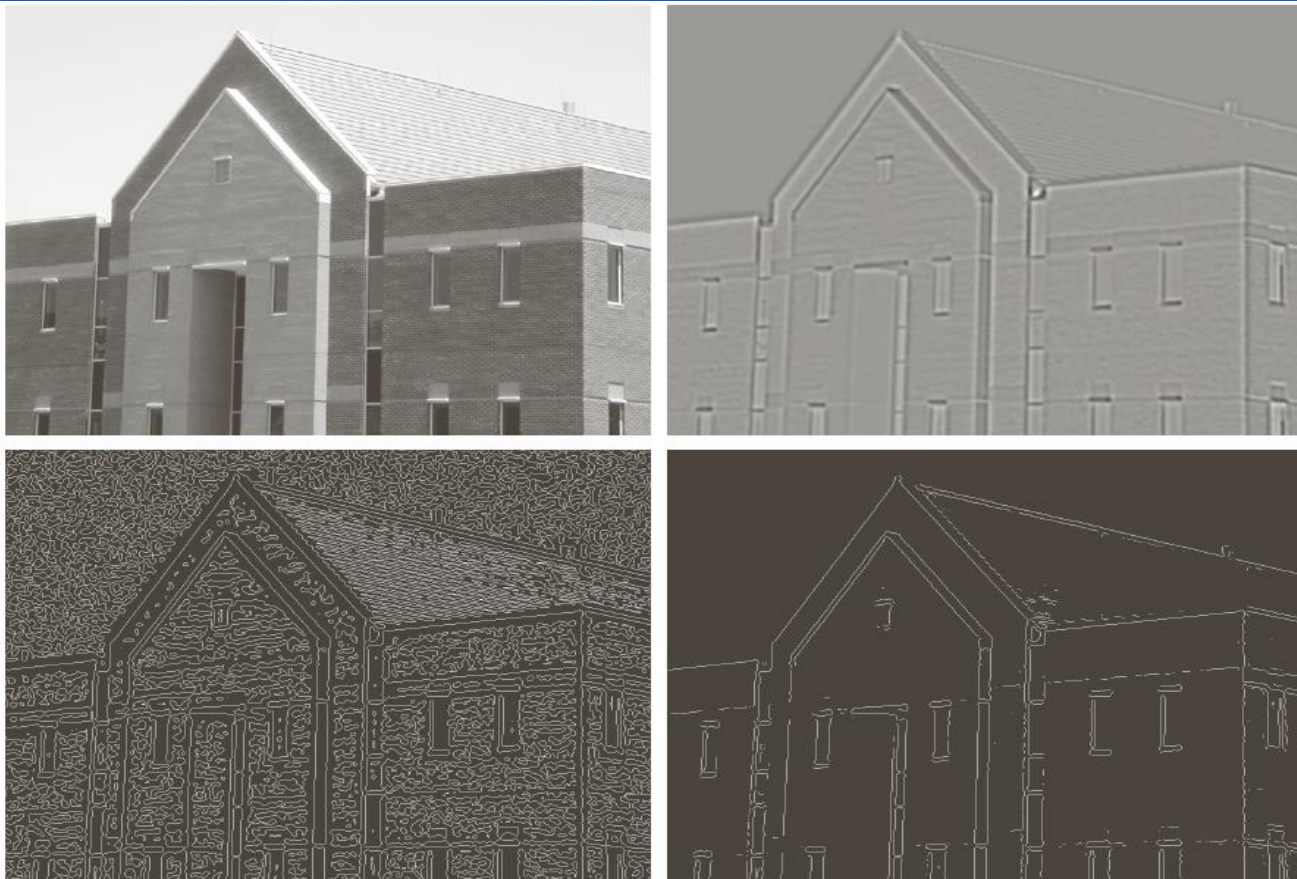


a b
c d

FIGURE 10.21

(a) Three-dimensional plot of the *negative* of the LoG. (b) Negative of the LoG displayed as an image. (c) Cross section of (a) showing zero crossings. (d) 5×5 mask approximation to the shape in (a). The negative of this mask would be used in practice.

0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0



a	b
c	d

FIGURE 10.22

(a) Original image of size 834×1114 pixels, with intensity values scaled to the range $[0, 1]$. (b) Results of Steps 1 and 2 of the Marr-Hildreth algorithm using $\sigma = 4$ and $n = 25$. (c) Zero crossings of (b) using a threshold of 0 (note the closed-loop edges). (d) Zero crossings found using a threshold equal to 4% of the maximum value of the image in (b). Note the thin edges.

Edge Linking and Boundary Detection

Edge detection typically is followed by linking algorithms designed to assemble edge pixels into meaningful edges and/or region boundaries

Three approaches to edge linking

- Local processing
- Regional processing
- Global processing

Three approaches to edge linking

- Local processing
- Regional processing
- Global processing

Local Processing

- Analyze the characteristics of pixels in a small neighborhood about every point (x,y) that has been declared an edge point
- All points that are similar according to predefined criteria are linked, forming an edge of pixels.

Establishing similarity:

- (1) the strength (magnitude) and
- (2) the direction of the gradient vector.

A pixel with coordinates (s,t) in S_{xy} is linked to the pixel at (x,y) if both magnitude and direction criteria are satisfied.





Local Processing: Steps (2)

3. Scan the rows of g and fill (set to 1) all gaps (sets of 0s) in each row that do not exceed a specified length, K .
4. To detect gaps in any other direction, rotate g by this angle and apply the horizontal scanning procedure in step 3.

Regional Processing

- **The location of regions of interest in an image are known or can be determined**
- **Polygonal approximations can capture the essential shape features of a region while keeping the representation of the boundary relatively simple**
- **Open or closed curve**
Open curve: a large distance between two consecutive points in the ordered sequence relative to the distance between other points

Regional Processing: Steps

1. Let P be the sequence of ordered, distinct, 1-valued points of a binary image. Specify two starting points, A and B .
2. Specify a threshold, T , and two empty stacks, **OPEN** and **CLOSED**.
3. If the points in P correspond to a closed curve, put A into **OPEN** and put B into **OPEN** and **CLOSES**. If the points correspond to an open curve, put A into **OPEN** and B into **CLOSED**.
4. Compute the parameters of the line passing from the last vertex in **CLOSED** to the last vertex in **OPEN**.

continue

5. Compute the distances from the line in Step 4 to all the points in P whose sequence places them between the vertices from Step 4. Select the point, $V_{\max}$, with the maximum distance, $D_{\max}$
6. If $D_{\max} > T$, place $V_{\max}$ at the end of the OPEN stack as a new vertex. Go to step 4.
7. Else, remove the last vertex from OPEN and insert it as the last vertex of CLOSED.
8. If OPEN is not empty, go to step 4.
9. Else, exit. The vertices in CLOSED are the vertices of the polygonal fit to the points in P .







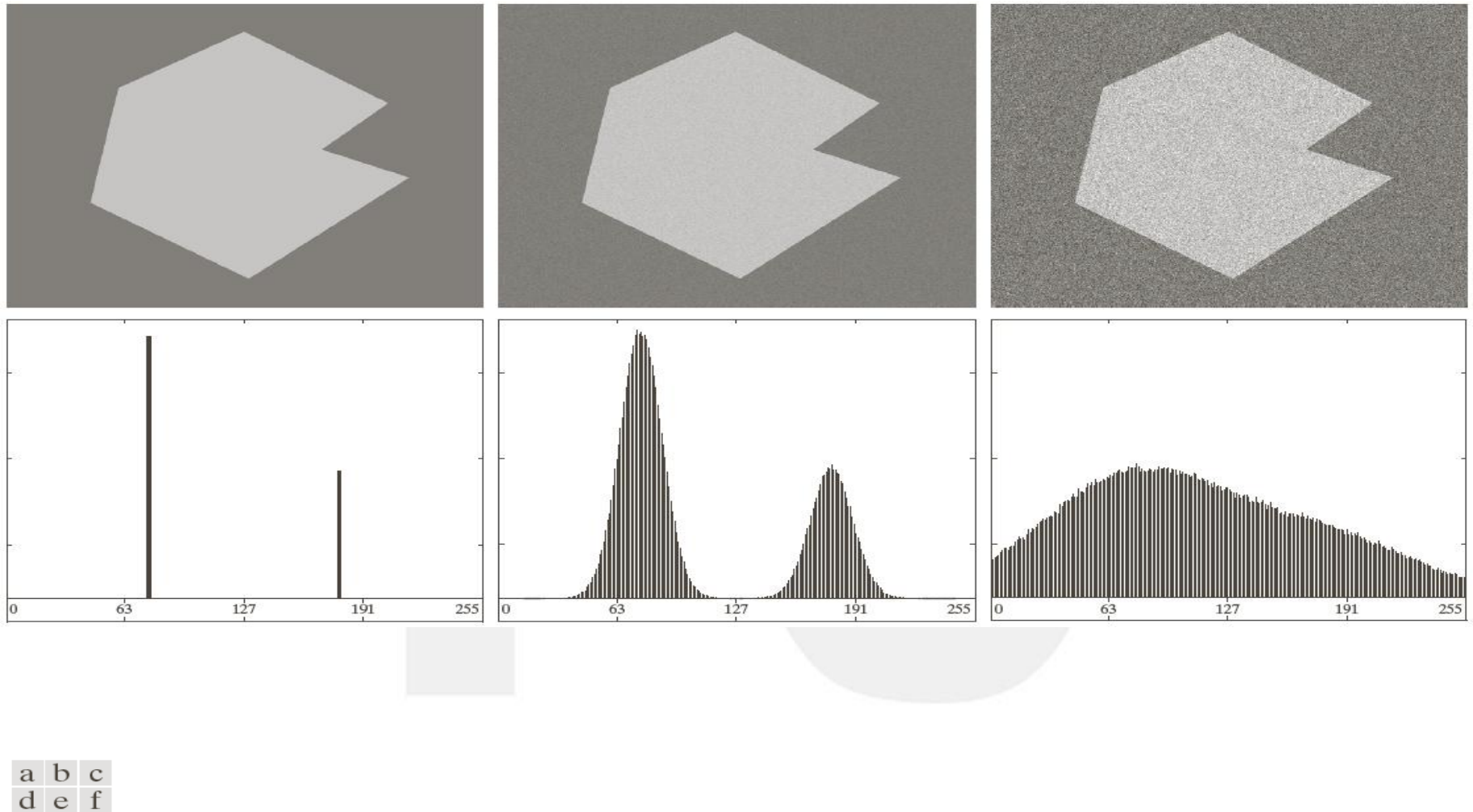
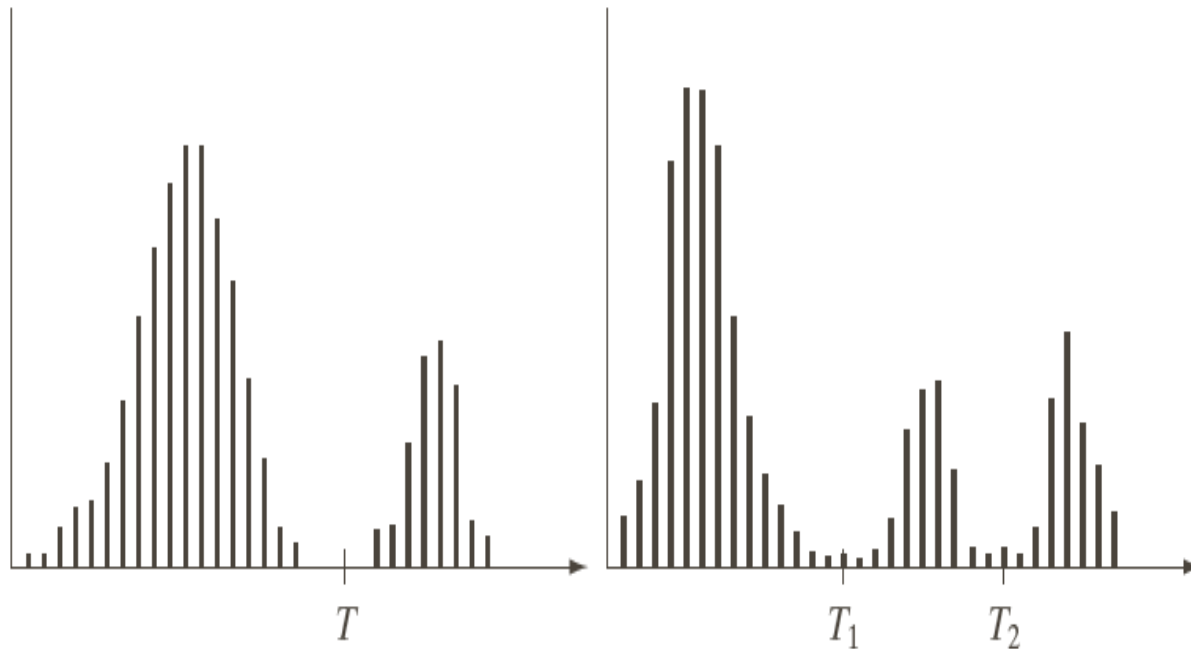


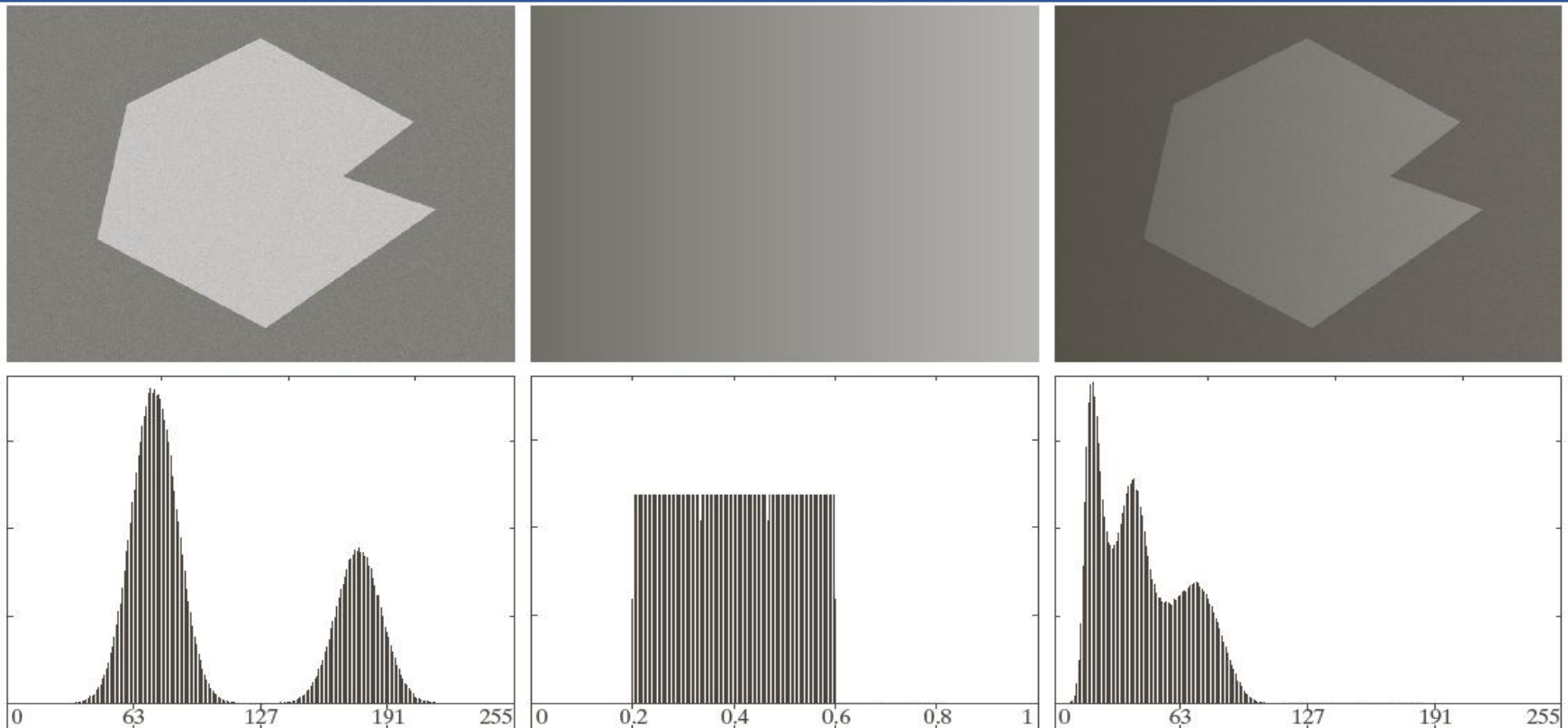
FIGURE 10.36 (a) Noiseless 8-bit image. (b) Image with additive Gaussian noise of mean 0 and standard deviation of 10 intensity levels. (c) Image with additive Gaussian noise of mean 0 and standard deviation of 50 intensity levels. (d)–(f) Corresponding histograms.



a b

FIGURE 10.35
Intensity
histograms that
can be partitioned
(a) by a single
threshold, and
(b) by dual
thresholds.

The Role of Illumination and Reflectance



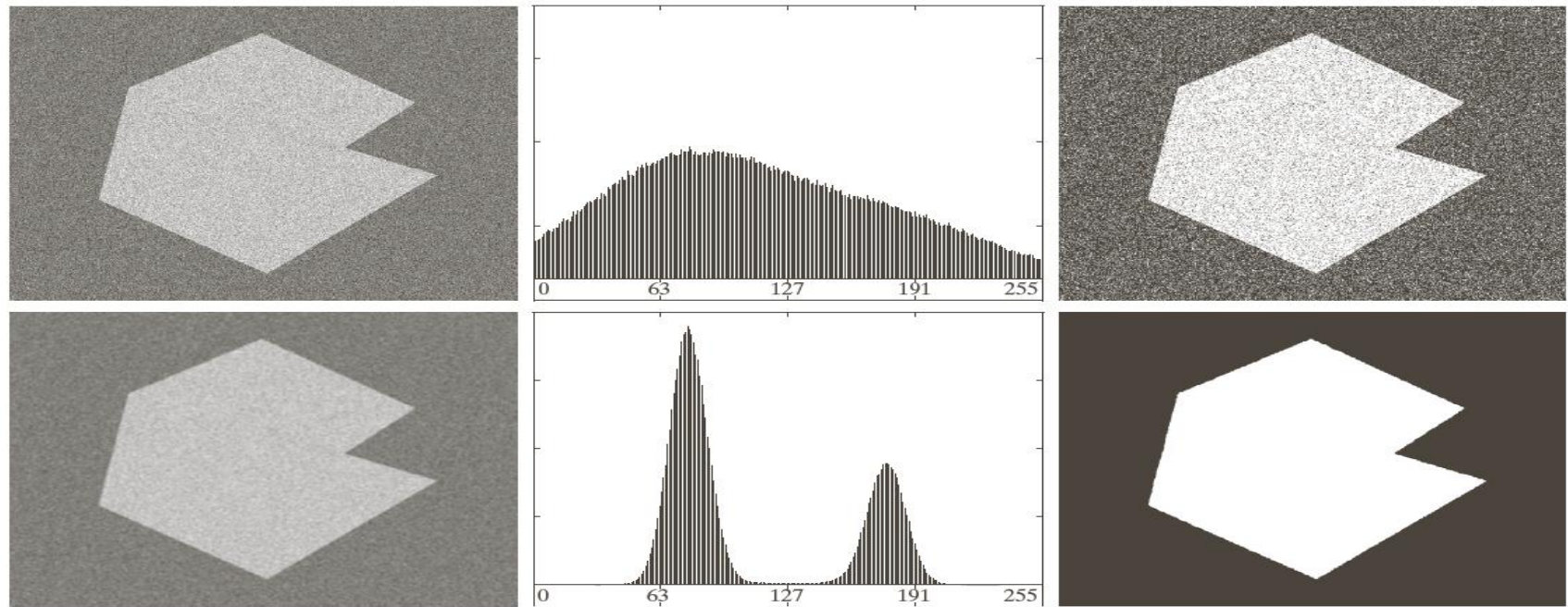
a	b	c
d	e	f

FIGURE 10.37 (a) Noisy image. (b) Intensity ramp in the range $[0.2, 0.6]$. (c) Product of (a) and (b). (d)–(f) Corresponding histograms.

Basic Global Thresholding

1. Select an initial estimate for the global threshold, T .
2. Segment the image using T . It will produce two groups of pixels: $G1$ consisting of all pixels with intensity values $> T$ and $G2$ consisting of pixels with values $\leq T$.
3. Compute the average intensity values $m1$ and $m2$ for the pixels in $G1$ and $G2$, respectively.
4. Compute a new threshold value.
5. Repeat Steps 2 through 4 until the difference between values of T in successive iterations is smaller than a predefined parameter ϵ .

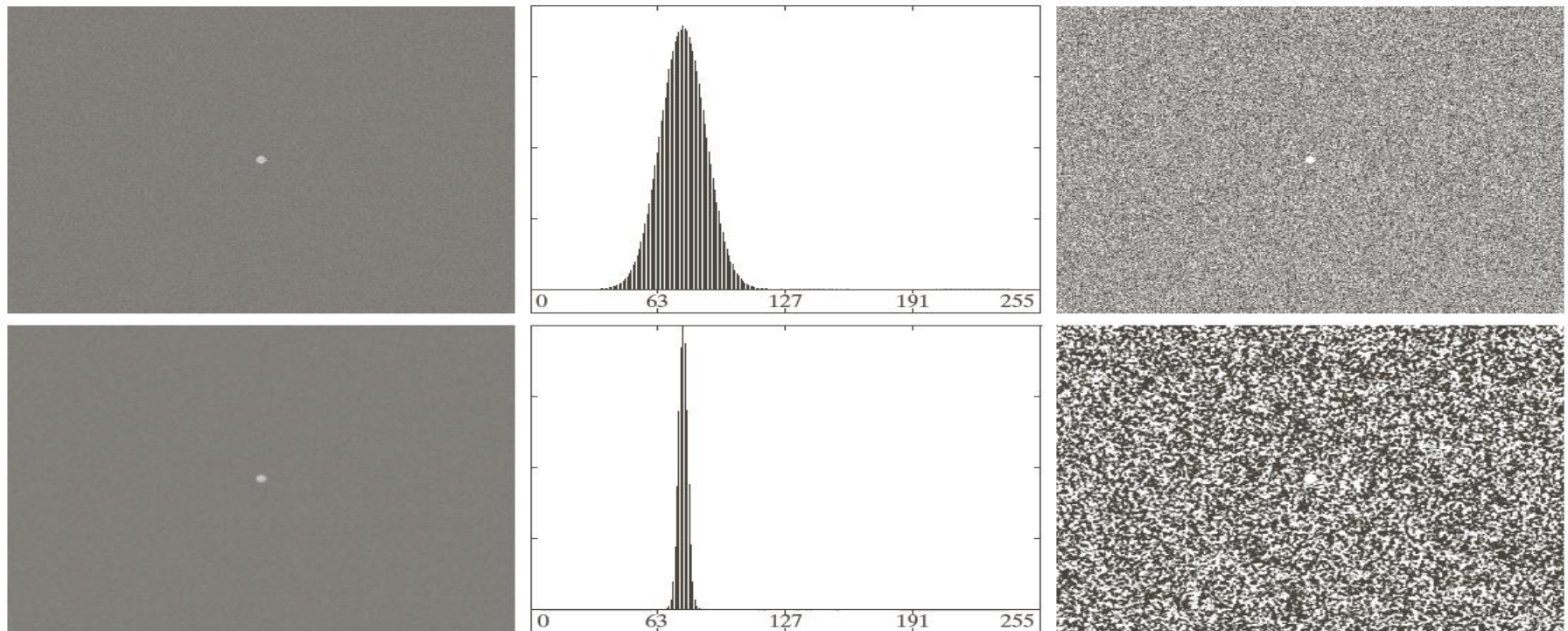
Using Image Smoothing to Improve Global Thresholding



a	b	c
d	e	f

FIGURE 10.40 (a) Noisy image from Fig. 10.36 and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5×5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method.

Using Edges to Improve Global Thresholding



a	b	c
d	e	f

FIGURE 10.41 (a) Noisy image and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5×5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method. Thresholding failed in both cases.

Using Edges to Improve Global Thresholding

1. Compute an edge image as either the magnitude of the gradient, or absolute value of the Laplacian of $f(x,y)$
2. Specify a threshold value T
3. Threshold the image and produce a binary image, which is used as a mask image; and select pixels from $f(x,y)$ corresponding to “strong” edge pixels
4. Compute a histogram using only the chosen pixels in $f(x,y)$
5. Use the histogram from step 4 to segment $f(x,y)$ globally

continue

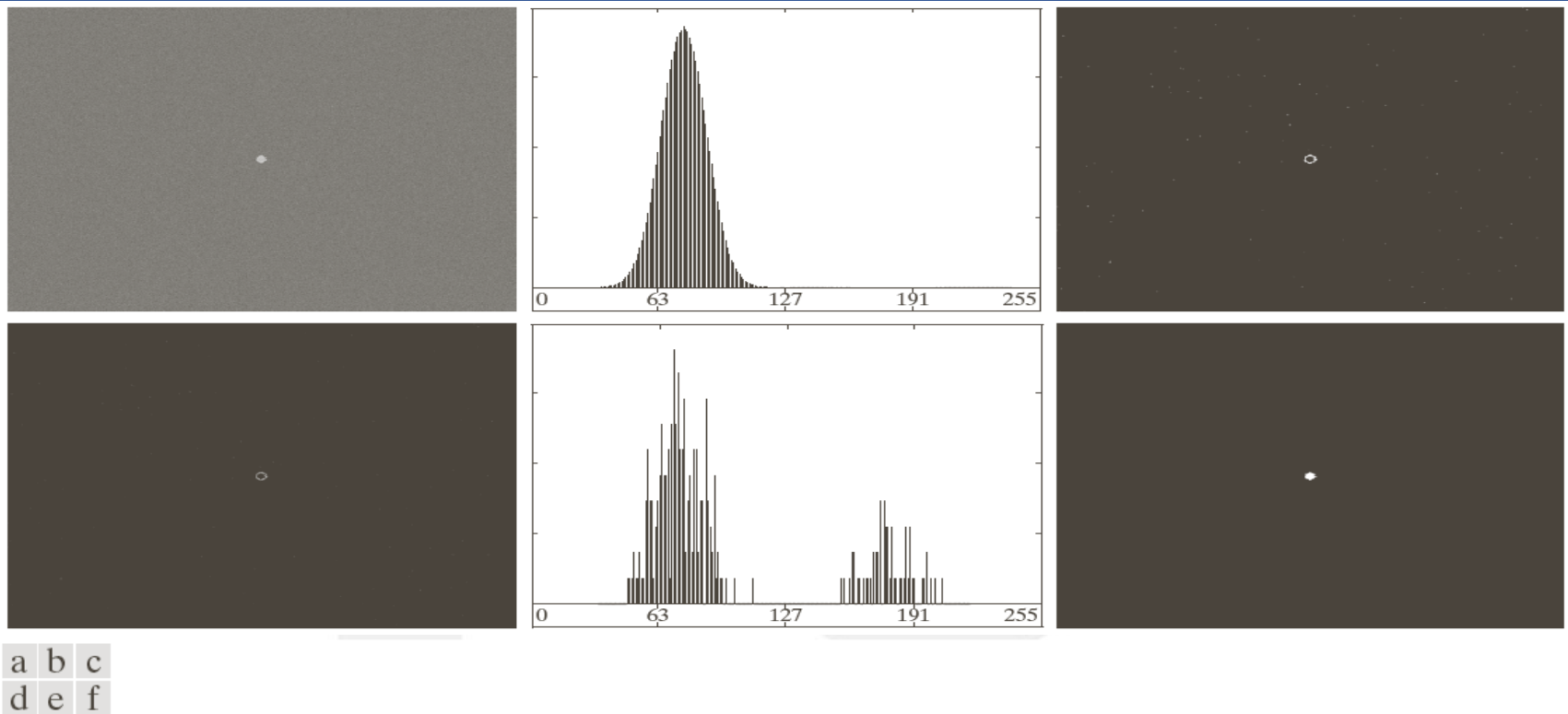
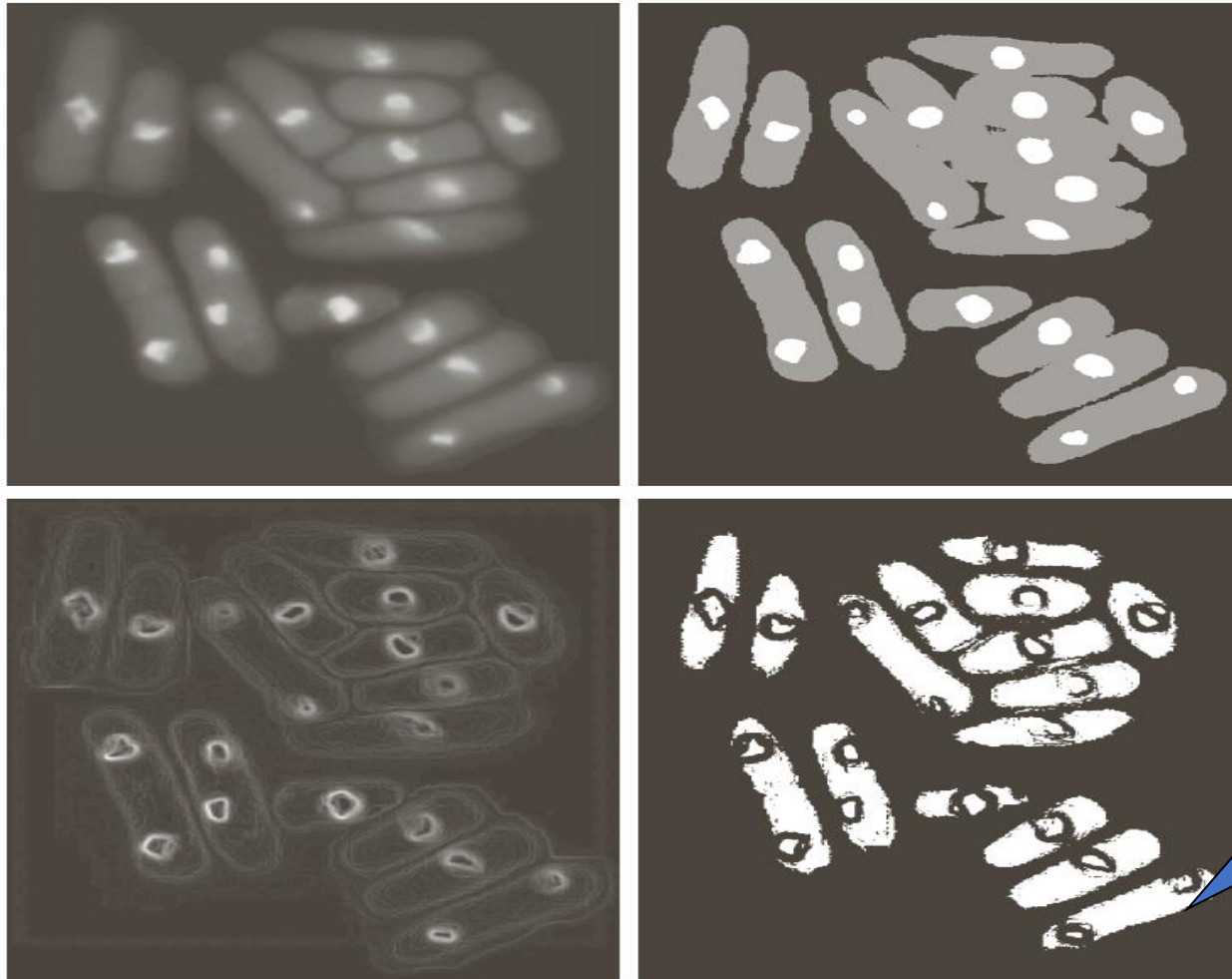


FIGURE 10.42 (a) Noisy image from Fig. 10.41(a) and (b) its histogram. (c) Gradient magnitude image thresholded at the 99.7 percentile. (d) Image formed as the product of (a) and (c). (e) Histogram of the nonzero pixels in the image in (d). (f) Result of segmenting image (a) with the Otsu threshold based on the histogram in (e). The threshold was 134, which is approximately midway between the peaks in this histogram.







a	b
c	d

FIGURE 10.48

(a) Image from Fig. 10.43.

(b) Image segmented using the dual thresholding approach discussed in Section 10.3.6.

(c) Image of local standard deviations.

(d) Result obtained using local thresholding.

$$a=30$$

$$b=1.5$$

$$m_{xy} = m_G$$

Region-Based Segmentation

- **Region Growing**
 1. Region growing is a procedure that groups pixels or subregions into larger regions.
 2. The simplest of these approaches is *pixel aggregation*, which starts with a set of “*seed*” points and from these grows regions by appending to each seed points those neighboring pixels that have similar properties (such as gray level, texture, color, shape).
 3. Region growing based techniques are better than the edge-based techniques in noisy images where edges are difficult to detect.



Region Growing based on 8-connectivity

-
-

PU

Suppose that we have the image given below.

(a) Use the region growing idea to segment the object. The seed for the object is the center of the image. Region is grown in horizontal and vertical directions, and when the difference between two pixel values is less than or equal to 5.

Table 1: Show the result of Part (a) on this figure.

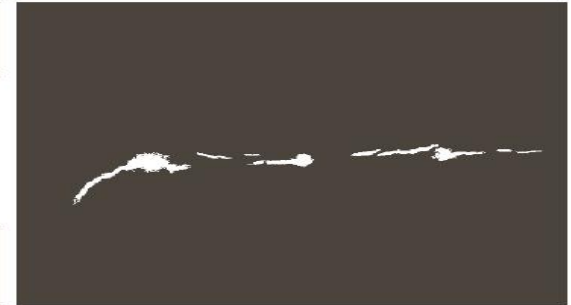
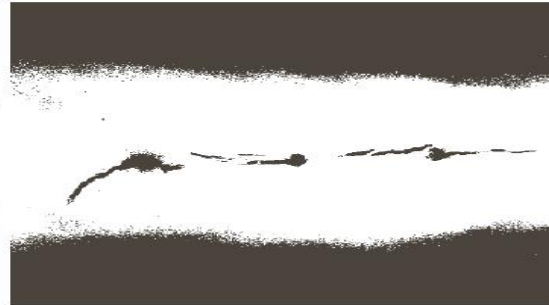
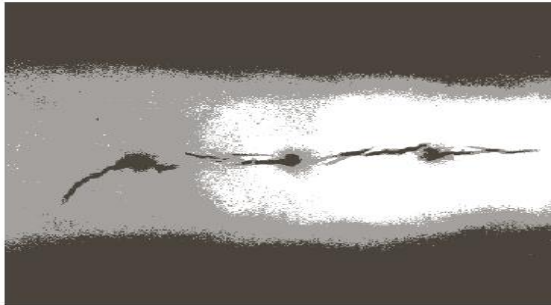
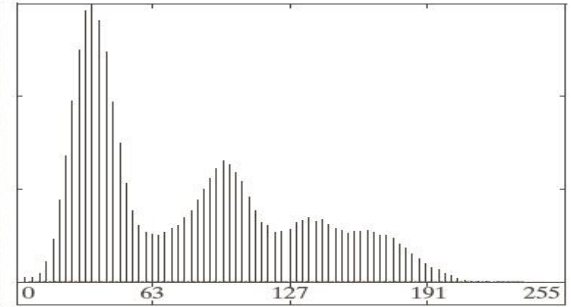
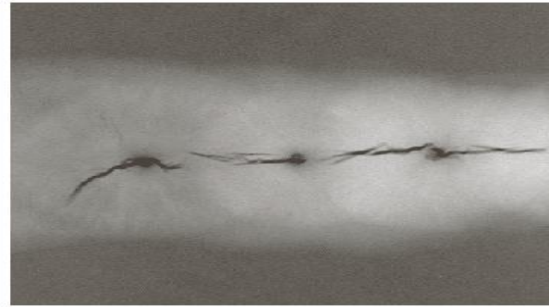
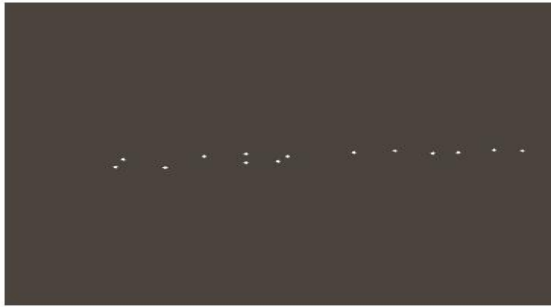
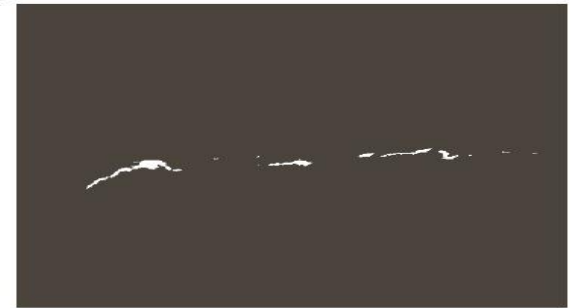
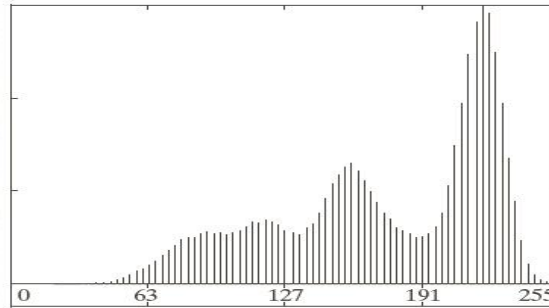
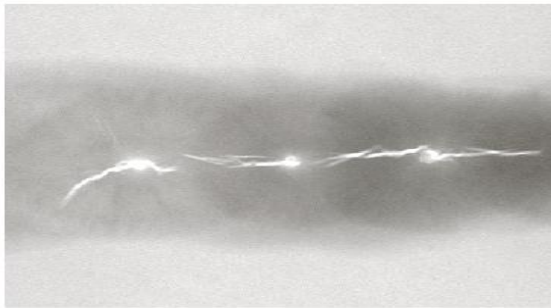
10	10	10	10	10	10	10
10	10	10	69	70	10	10
59	10	60	64	59	56	60
10	59	10	<u>60</u>	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10

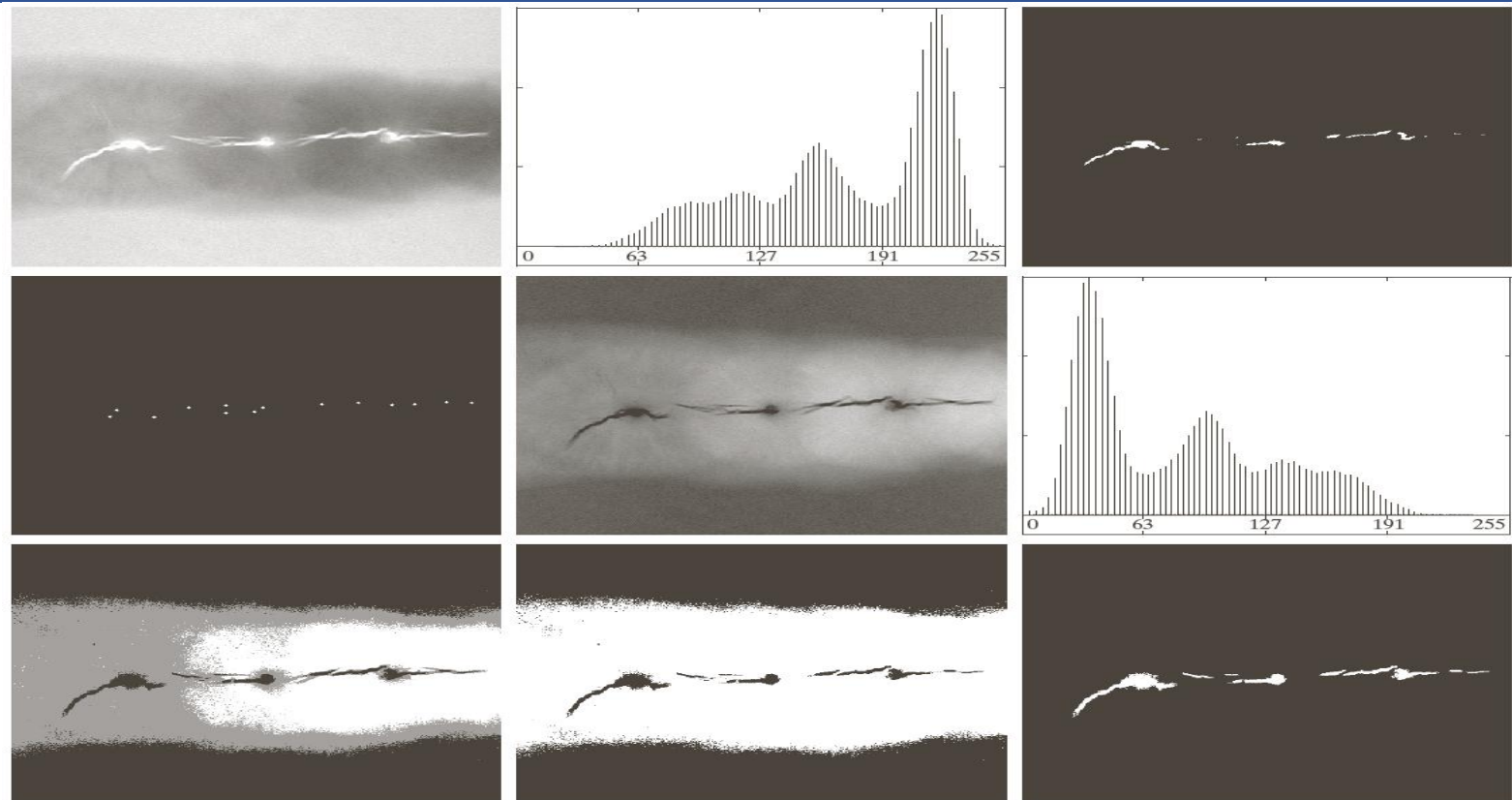
Suppose that we have the image given below.

(a) Use the region growing idea to segment the object. The seed for the object is the center of the image. Region is grown in horizontal and vertical directions, and when the difference between two pixel values is less than or equal to 5.

Table 1: Show the result of Part (a) on this figure.

10	10	10	10	10	10	10
10	10	10	69	70	10	10
59	10	60	64	59	56	60
10	59	10	<u>60</u>	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10

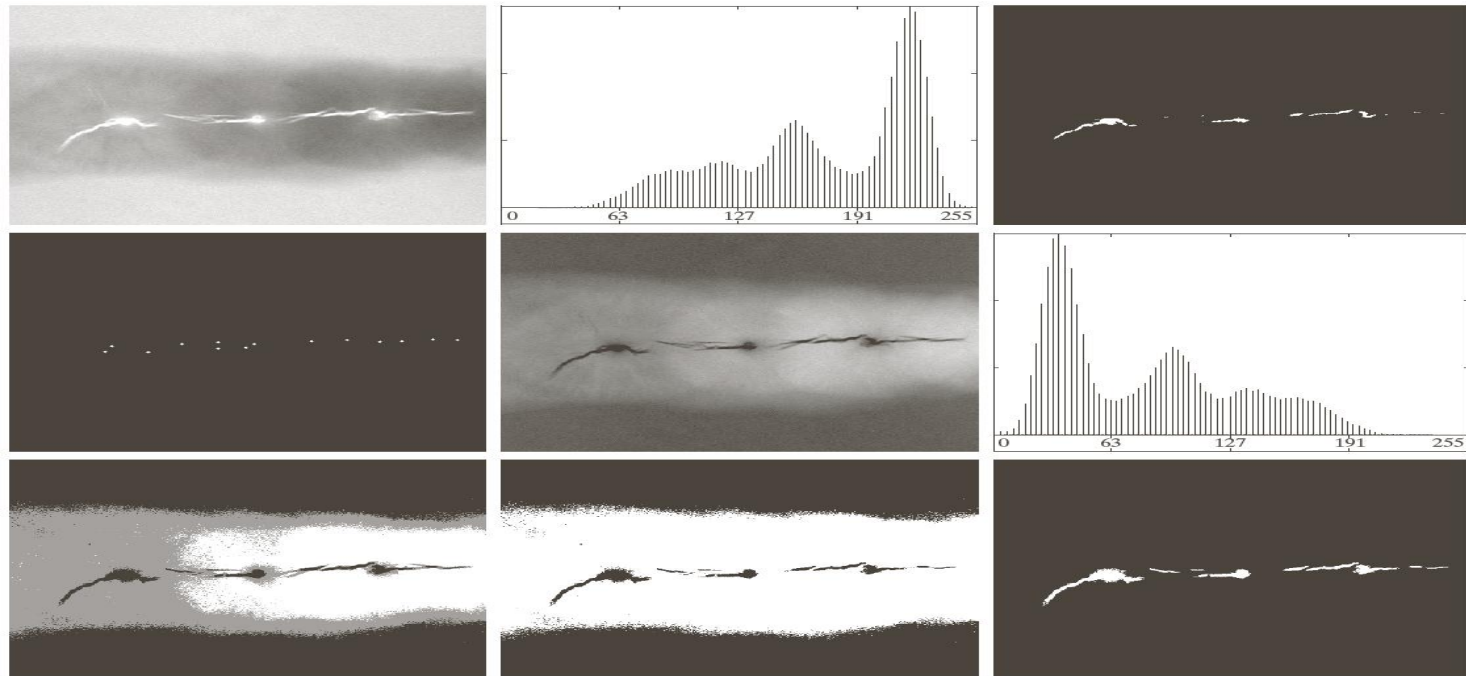




a	b	c
d	e	f
g	h	i

FIGURE 10.51 (a) X-ray image of a defective weld. (b) Histogram. (c) Initial seed image. (d) Final seed image (the points were enlarged for clarity). (e) Absolute value of the difference between (a) and (c). (f) Histogram of (e). (g) Difference image thresholded using dual thresholds. (h) Difference image thresholded with the smallest of the dual thresholds. (i) Segmentation result obtained by region growing. (Original image courtesy of X-TEK Systems, Ltd.)





a	b	c
d	e	f
g	h	i

FIGURE 10.51 (a) X-ray image of a defective weld. (b) Histogram. (c) Initial seed image. (d) Final seed image (the points were enlarged for clarity). (e) Absolute value of the difference between (a) and (c). (f) Histogram of (e). (g) Difference image thresholded using dual thresholds. (h) Difference image thresholded with the smallest of the dual thresholds. (i) Segmentation result obtained by region growing. (Original image courtesy of X-TEK Systems, Ltd.)

References

1. Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, 3rd Edition Pearson Education Publisher.
2. Pratt William, *Digital Image Processing*, John Wiley & Sons
3. S Jayaraman, S Esakirajan and T Verakumar *Digital Image Processing*, McGraw Hill
4. John C. Russ, *The Image Processing Handbook*, by; CRC Press, 2007
5. A. K. Jain, *Fundamentals of Digital Image Processing*, Pearson Education
6. Mark Nixon and Alberto Aguado, *Feature Extraction and Image Processing*, Academic Press, 2008



Parul[®] University

